

DATA ANALYSIS REPORT

Titanic Survival Analysis

An Exploratory Data Analysis of Passenger Survival Factors

Dataset	Titanic Passenger Dataset — Kaggle
Tools Used	Python (Pandas), PostgreSQL, Tableau
Records	891 passengers · 12 variables
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Executive Summary

This report presents a comprehensive analysis of passenger survival data from the RMS Titanic disaster. Using a structured data pipeline encompassing Python-based data preparation, PostgreSQL analysis, and Tableau visualisation, this study examines the demographic, socioeconomic, and logistical factors that influenced survival outcomes.

The dataset comprises 891 passenger records across 12 variables. Key findings reveal that overall survival rate stood at 38.38%, with survival outcomes strongly correlated to gender, passenger class, age group, fare paid, and family size.

1. Project Overview

The Titanic disaster remains one of the most studied historical events in modern data science. This project applies exploratory data analysis (EDA) and structured query techniques to the Titanic passenger dataset to uncover statistically meaningful patterns in survival outcomes.

The analysis investigates the following factors:

- Passenger demographics : gender, age group, and family composition
- Socioeconomic indicators : ticket class, cabin assignment, and fare paid
- Logistical variables : embarkation port and family size

The project follows a full data analytics workflow: data ingestion, cleaning, transformation, database storage, SQL-based analysis, and visual reporting via Tableau.

2. Dataset Summary

2.1 Dataset Information

Attribute	Details
Dataset Name	Titanic Passenger Dataset
Data Source	Kaggle Titanic Competition
Total Records (Rows)	891
Total Variables (Columns)	12

2.2 Feature Descriptions

The dataset contains 12 variables covering passenger identity, demographics, ticket information, and survival status:

Variable	Description	Type
PassengerId	Unique identifier for each passenger	Integer
Survived	Survival indicator (0 = No, 1 = Yes)	Binary
Pclass	Ticket class (1st, 2nd, or 3rd)	Ordinal

Variable	Description	Type
Name	Full name of the passenger	String
Sex	Gender of the passenger	Categorical
Age	Age in years	Numeric
SibSp	Number of siblings / spouses aboard	Integer
Parch	Number of parents / children aboard	Integer
Ticket	Ticket number	String
Fare	Passenger fare paid (£)	Numeric
Cabin	Cabin number (where recorded)	String
Embarked	Port of embarkation (S / C / Q)	Categorical

2.3 Dataset Preview

The table below shows a snapshot of the first five rows of the dataset as loaded into Python:

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Figure A: Dataset preview — df.head()

2.4 Initial Data Quality Assessment

A preliminary quality assessment identified missing values in three columns:

Variable	Missing Values	Handling Strategy
Age	177 records	Replaced with median age
Cabin	687 records	Converted to binary indicator (Cabin_known)
Embarked	2 records	Replaced with most frequent port (mode)

3. Data Preparation and Cleaning

All data preparation was performed in Python using the Pandas library across four sequential stages: initial exploration, missing value handling, data transformation, and feature engineering.

3.1 Initial Exploration

The dataset was imported into Python and examined using the following diagnostic methods:

- `df.head()` — preview of the first five records
- `df.info()` — column data types and non-null counts
- `df.describe()` — descriptive statistics for numeric columns
- `df.isnull().sum()` — identification of missing values per column

Dataset Structure — `df.info()`

The output below confirms column data types and non-null value counts across all 12 variables:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId     891 non-null    int64
1   Survived        891 non-null    int64
2   Pclass         891 non-null    int64
3   Name           891 non-null    object
4   Sex            891 non-null    object
5   Age            714 non-null    float64
6   SibSp          891 non-null    int64
7   Parch          891 non-null    int64
8   Ticket         891 non-null    object
9   Fare           891 non-null    float64
10  Cabin          204 non-null    object
11  Embarked       889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

Figure B: Dataset structure — `df.info()`

Descriptive Statistics — `df.describe()`

Summary statistics for all numeric columns confirm ranges, means, and standard deviations:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare	Cabin_known
count	891.000000	891.000000	891.000000	891.000000	891.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.361582	0.523008	0.381594	32.204208	0.228956
std	257.353842	0.486592	0.836071	13.019697	1.102743	0.806057	49.693429	0.420397
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	22.000000	0.000000	0.000000	7.910400	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200	0.000000
75%	668.500000	1.000000	3.000000	35.000000	1.000000	0.000000	31.000000	0.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200	1.000000

Figure C: Descriptive statistics — `df.describe()`

3.2 Missing Value Handling

Age Column

Missing age values were imputed using the median to minimise the influence of outliers on the distribution.

```
df["Age"].fillna(df["Age"].median(), inplace=True)
```

Embarked Column

With only two missing entries, the Embarked column was imputed using the mode (most frequently occurring embarkation port).

```
df["Embarked"].fillna(df["Embarked"].mode()[0], inplace=True)
```

Cabin Column

Given the high proportion of missing cabin records (77%), a binary indicator variable was created to preserve the informational value of cabin availability:

```
df["Cabin_known"] = df["Cabin"].notna().astype(int)
```

Missing Values After Cleaning

The following figure confirms that all missing values were resolved post-cleaning:

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	0
Cabin_known	0
dtype:	int64

Figure D: Missing values after cleaning

3.3 Data Transformation

Embarkation Port Standardisation

Single-character embarkation codes were mapped to full port names for readability:

Original Code	Full Port Name
S	Southampton
C	Cherbourg
Q	Queenstown

```
df["Embarked"] = df["Embarked"].replace({
    "S": "Southampton",
    "C": "Cherbourg",
    "Q": "Queenstown"
})
```

3.4 Feature Engineering

Age Group Classification

A derived categorical variable, `Age_group`, was created by binning passenger ages into five demographic cohorts:

Age Range	Category Label
0 – 12	Child
13 – 19	Teenager
20 – 35	Young Adult
36 – 55	Adult
56 – 80	Senior

```
bins    = [0, 12, 19, 35, 55, 80]
labels  = ['Child', 'Teenager', 'Young-Adult', 'Adult', 'Senior']

df['Age_group'] = pd.cut(
    df['Age'],
    bins=bins,
    labels=labels,
    include_lowest=True
)
```

Engineered Dataset — Age Group Preview

The figure below shows the `Age_group` column successfully added to the dataset:

	Age	Age_group
0	22.0	Young-Adult
1	38.0	Adult
2	26.0	Young-Adult
3	35.0	Young-Adult
4	35.0	Young-Adult

Figure E: Age_group feature in the engineered dataset

4. Database Integration

Following the completion of all data preparation steps in Python, the cleaned and enriched dataset was loaded into a PostgreSQL relational database. This enabled the use of structured SQL queries for efficient aggregation, filtering, and cross-tabulation of passenger records.

5. Data Analysis Using SQL

Ten structured SQL queries were executed against the PostgreSQL database to address the core analytical questions of this study. Each query targeted a specific dimension of the survival problem, and all results are presented with their corresponding output screenshots below.

5.1 SQL Query Analysis

1. Overall Survival Rate

This query determined the baseline percentage of passengers who survived the disaster. The result establishes the foundation for all subsequent comparative analyses.

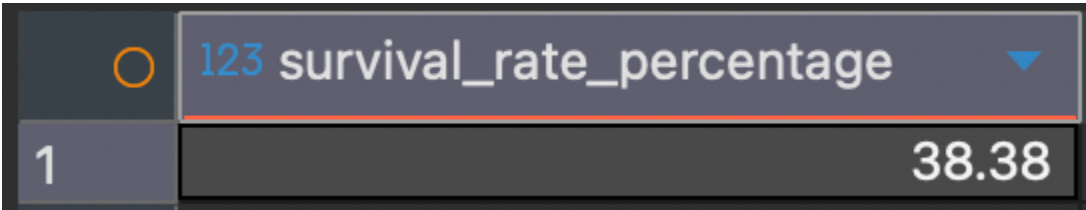


Figure 1: Overall Survival Rate

2. Survival by Gender

This query assessed whether gender played a statistically significant role in determining passenger survival rates.

	A-Z Sex	123 total_passengers	123 survivors	123 survival_rate
1	female	314	233	74.2
2	male	577	109	18.89

Figure 2: Survival by Gender

3. Survival by Passenger Class

This query investigated whether passengers travelling in different ticket classes experienced materially different survival rates.

	123 Pclass	123 total_passengers	123 survivors	123 survival_rate
1	1	216	136	62.96
2	2	184	87	47.28
3	3	491	119	24.24

Figure 3: Survival by Passenger Class

4. Survival by Age Group

This query examined survival patterns across the five engineered age cohorts, identifying those with the highest and lowest survival rates.

	A-Z Age_group	123 total_passengers	123 survivors	123 survival_rate
1	Child	69	40	57.97
2	Teenager	95	39	41.05
3	Adult	177	71	40.11
4	Young-Adult	510	180	35.29
5	Senior	40	12	30

Figure 4: Survival by Age Group

5. Survival by Embarkation Port

This query analysed whether passengers boarding from different embarkation ports exhibited varying chances of survival.

	A-Z mbarked	123 total_passengers	123 survivors	123 survival_rate
1	Cherbourg	168	93	55.36
2	Queenstown	77	30	38.96
3	Southampton	646	219	33.9

Figure 5: Survival by Embarkation Port

6. Family Size and Survival

This query determined whether passengers travelling alone, with a small family, or with a large family experienced different survival outcomes.

	A-Z family_size_group	123 total_passengers	123 survival_rate
1	Small Family	292	57.88
2	Alone	537	30.35
3	Large Family	62	16.13

Figure 6: Family Size and Survival

7. Fare and Survival

This query investigated whether passengers who paid higher fares were more likely to survive than those who paid lower fares.

	A-Z fare_group	123 total_passengers	123 survival_rate
1	High	222	58.11
2	High-Mid	229	44.54
3	Low-Mid	217	30.88
4	Low	223	19.73

Figure 7: Fare and Survival

8. Cabin Information and Survival

This query examined whether passengers with recorded cabin information exhibited higher survival rates than those without cabin records.

	123 Cabin_known	123 total_passengers	123 survivors	123 survival_rate
1	1	204	136	66.67
2	0	687	206	29.99

Figure 8: Cabin Information and Survival

9. Combined Effect of Gender and Passenger Class

This query evaluated how survival rates varied when both gender and passenger class were considered simultaneously, revealing interaction effects between the two variables.

	A-Z Sex	123 Pclass	123 total_passengers	123 survival_rate
1	female	1	94	96.81
2	female	2	76	92.11
3	female	3	144	50
4	male	1	122	36.89
5	male	2	108	15.74
6	male	3	347	13.54

Figure 9: Combined Effect of Gender and Passenger Class

10. Highest-Risk Passenger Groups

This query identified passenger segments with the lowest observed survival rates by analysing combinations of gender, passenger class, and age group.

	AZ Sex	123 Pclass	AZ Age_group	123 total_passengers	123 survival_rate
1	male	3	Senior	6	0
2	female	2	Senior	1	0
3	female	1	Child	1	0
4	male	2	Adult	24	4.17
5	male	3	Adult	42	7.14
6	male	3	Teenager	38	7.89
7	male	2	Young-Adult	59	8.47
8	male	2	Teenager	10	10
9	male	3	Young-Adult	236	13.56
10	female	3	Adult	14	14.29

Figure 10: Highest-Risk Passenger Groups

6. Tableau Dashboard

An interactive Tableau dashboard was developed to complement the SQL analysis, enabling dynamic exploration of survival patterns across multiple dimensions simultaneously.

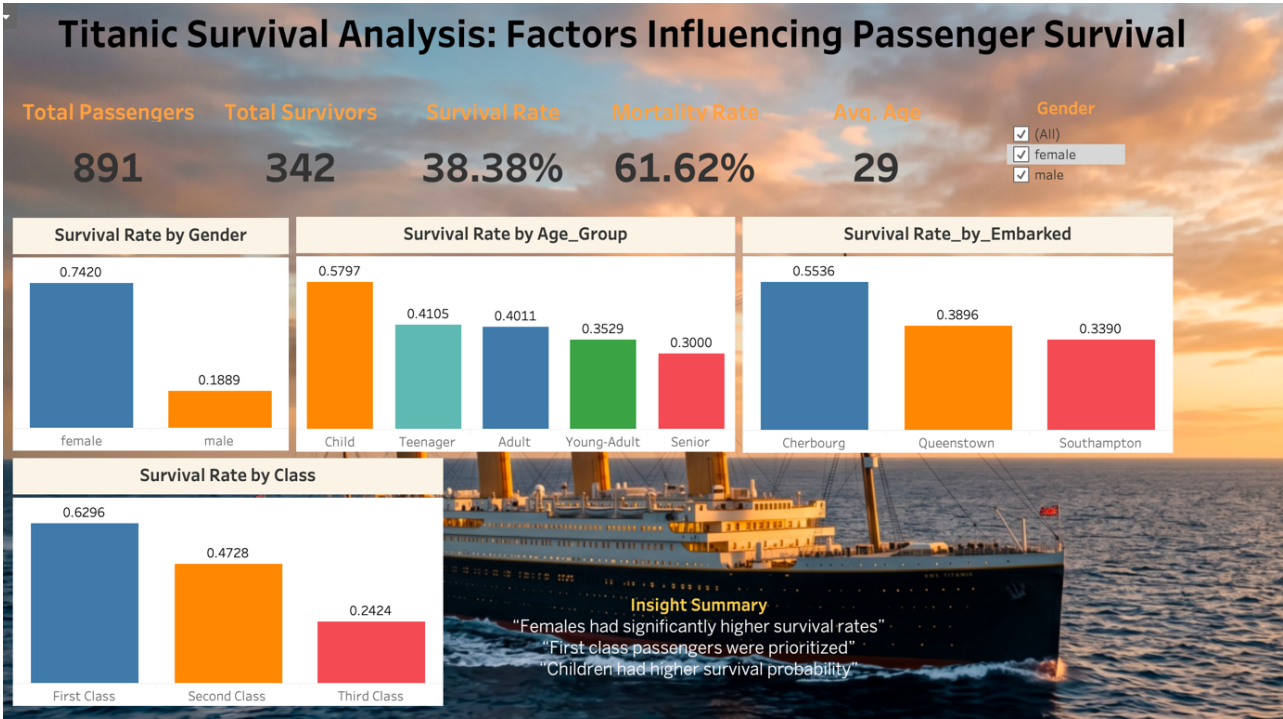


Figure F: Titanic Survival Analysis — Tableau Dashboard

The dashboard features the following components:

- Overall survival rate summary card
- Survival breakdown by gender, passenger class, and age group

- Embarkation port analysis with comparative survival rates
- Age group and its relationship to survival outcomes

7. Key Findings and Insights

The analysis produced ten principal findings, each illuminating a distinct dimension of the survival patterns observed across the Titanic passenger manifest.

1	Overall Survival Rate Was Relatively Low Only 38.38% of passengers survived the Titanic disaster. This baseline figure contextualises all subsequent group-level comparisons and confirms that the majority did not survive.
2	Female Passengers Had Significantly Higher Survival Rates Female passengers exhibited a substantially higher survival rate than male passengers, consistent with the maritime convention of prioritising women and children during evacuation procedures.
3	First-Class Passengers Were More Likely to Survive Passengers in First Class recorded the highest survival rate across all ticket classes, while Third-Class passengers experienced the lowest. Socioeconomic status influenced access to lifeboats and emergency assistance.
4	Age Was Associated with Survival Outcomes Children (ages 0–12) achieved the highest survival rate among all age cohorts, while Senior passengers (ages 56–80) recorded the lowest. Age likely influenced evacuation priority during the disaster.
5	Embarkation Port Correlated with Survival Differences Passengers boarding at Cherbourg demonstrated the highest survival rate, while those embarking from Southampton had the lowest. This variation likely reflects underlying differences in class distributions across ports.
6	Small Family Groups Had Better Survival Outcomes Passengers travelling with small family groups fared better than those travelling alone or in large family units. Moderate family support appears to have been advantageous during the evacuation.
7	Higher Fares Were Associated with Greater Survival Likelihood Passengers who paid higher ticket fares were generally more likely to survive, consistent with the higher survival rates observed among First-Class passengers who typically paid premium fares.
8	Cabin Assignment Was a Strong Survival Indicator

Passengers with recorded cabin information exhibited significantly higher survival rates. Cabin assignment was predominantly a feature of First-Class accommodation, reinforcing the link between class and survival.

9

Gender and Passenger Class Together Strongly Determined Survival

First-Class females showed the highest survival rates of any subgroup, while Third-Class males experienced the lowest. The joint effect of social status and gender was more predictive than either variable alone.

10

Certain Passenger Segments Faced Extremely Low Survival Chances

Several groups with exceptionally low survival rates were identified — particularly male passengers in Third Class across older age cohorts. These segments represented the most vulnerable populations during the disaster.

8. Conclusion

This analysis demonstrates that Titanic passenger survival was shaped by a convergence of social, economic, and demographic factors. The data consistently shows that female passengers in higher ticket classes, particularly those with cabin assignments and moderate family support , faced markedly better odds of survival.

Conversely, Third-Class male passengers, particularly older individuals travelling alone or in large family groups, constituted the most at-risk population. The fare paid, serving as a proxy for socioeconomic status, proved to be among the most consistent predictors of survival across all analyses.